Basin Escape, Player-0 Activation, and Settlement Preorders

A formal note on "White wins," Eureka, IAE, chess, static proofs, and passwords

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Central interpretation. When a conjecture settler receives *"White wins,"* the intended first-person conjecture is *"I will prove the theorem"*; in GBG language, *"I will reach the target."* This is a prediction about a designated settler's future verified achievement, *not* a command, a generic capability assertion, or the mere existence of some possible route.

Abstract

Consider an automated prover stalled in a search basin: it is unsure what to do next, just as a chess player may be unsure how to move. We formalize three proposed escape methods: (i) a Player-0-activated AMLD+0 federation investigating the first-person conjecture "White wins," (ii) Eureka-style search for nontrivial intermediary lemmas, and (iii) Iterative Adaptive Eureka (IAE) search that revises these lemma-discovery attempts. Verified products enter a provenance-preserving shared BANK, and simulation-isomorphism classes (SICs) are ordered by an explicitly certificate-preserving simulation preorder. Under stated assumptions, settlement regions are upper sets and, in a well-founded quotient order, admit minimal-antichain generators. We separate three importantly different environments: chess has a *moving adversary*; a fixed formal theory and theorem target do *not*; a password may be fixed or may change in an adaptive challenge system. We additionally distinguish finite ordinary chess from a GBG+/formal search universe with potentially infinitely many proof targets and obstacles. All speedup claims remain hypotheses requiring resource-accounted experiments.

1 Motivation: not knowing the next move

Imagine a person studying a chess position and unable to decide which legal move to make. This uncertainty does not imply that there is no winning move. The difficulty can be in the *search policy*: poor move ordering, a missing tactic, an unexamined defensive branch, or an insufficiently useful intermediate position. We call a region of search states a *quicksand basin* when substantial charged effort yields little or no certified advancement under a declared progress metric.

A basin is a property of the search and the chosen diagnostic, not automatically of the mathematical target. A stalled search does not prove impossibility, falsity, or independence. The Grass-Growth Principle in the accompanying Quicksand manuscript calls for measuring progress as well as nonprogress, and for distinguishing useful new knowledge from mere movement through states [2].

The critical asymmetry. In *chess*, Black has legal responses that alter the board. In a *fixed formal system*, the axioms and the proposition being sought remain fixed; what changes is the ATP's internal proof state, available lemmas, and search strategy. In a *fixed-password* challenge, the target secret likewise does not move; in rotating-password or adaptive-challenge settings, the target or acceptance rule may change. These environments must not be silently identified.

2 Three target settings, one search vocabulary

Definition 2.1 (Location report and rule-complete chess state)**.** *Let A be the set of board squares, P the set of occupant tokens, and E the distinguished* empty-square token*. A location report is*

$$f \in [A \to (P \cup \{E\})].$$

A rule-complete chess state is $s = (f, \tau, \eta)$, where τ identifies whose turn it is and η records additional rule-relevant data (for example castling, en passant, and draw-related state). Legal chess moves form a partial transition relation $s \xrightarrow{\mu} s'$.

The symbol E, not the empty-set symbol, denotes an empty square. Two distinct reported states need not by themselves certify a legal move; legality depends on an authenticated transition and the full rule-complete state.

Definition 2.2 (Chess strategy claim)**.** *For an admissible starting state s, write* $\mathsf{ForceWin}(s)$ *for the statement that White has a legal strategy that reaches a declared White-winning terminal condition against* every *legal Black strategy (subject to a clearly stated termination/horizon convention):*

$$\mathsf{ForceWin}(s) \;:\Longleftrightarrow\; \exists \sigma_W \, \forall \sigma_B \quad \mathsf{Goal}\big(\mathsf{Play}(s, \sigma_W, \sigma_B)\big).$$

A certificate generally requires a strategy tree, not one favorable path.

Definition 2.3 (Static formal target)**.** *A static theorem-proving instance is $q = (T, c, V)$ where T is a fixed theory, c a fixed target formula, and $V(\pi, T \vdash c)$ is a sound proof-checking relation. The search may evolve through states $z_0, z_1, \ldots$, but T and c do not change merely because an agent searches. A certificate is an accepted finite derivation π.*

Definition 2.4 (Password target)**.** *A fixed-password task has a hidden secret w and a fixed acceptance predicate* $\mathsf{Accept}(x, w)$*. Guessing changes the search history, not w. An adaptive-password task instead permits a legal evolution (w_t, Accept_t), e.g. rotation, challenge updates, or reactions to attempts. It is only in the latter setting that an adversary-like moving target is present.*

Proposition 2.5 (Path versus strategy)**.** *A verified chess path to a White win need not prove* $\mathsf{ForceWin}(s)$*. A verified path to a formal proof certificate* does *establish its stated static theorem, provided all inference dependencies are valid.*

Proof. A chess path fixes Black's replies; another legal Black reply may defeat the chosen continuation. The forced-win claim quantifies over all Black replies. By contrast, in a sound static proof calculus an accepted finite derivation already witnesses $T \vdash c$; there is no subsequent legal adversarial move that changes the fixed entailment fact. □

Therefore. "Black moves around" is literal in chess, optional in an adaptive-password game, and *not* literal in ordinary theorem proving. Counterexample search or refutation agents do not change the original axioms or theorem target by acting as an adversary; they examine consequences of the same fixed problem.

Definition 2.6 (Families of theorem targets and obstacles)**.** *Fix a formal environment T and a candidate family of proof targets*

$$\mathcal{T} = \{t_i : i \in I\}, \qquad \mathcal{O} = \{o_j : j \in J\},$$

where either index set may be finite or infinite. The members of $\mathcal{O}$ represent proof obligations, unresolved cases, refutation challenges, or other explicitly typed impediments, not mobile chess pieces. A single frozen experiment chooses an original target $c \in \mathcal{T}$; the existence of many other possible theorem targets does not authorize changing c mid-run. A successful formally checked derivation of one t_i can be a useful intermediate lemma even when it is not the frozen target c.

Remark 2.7 (Many possible "checkmates" are not many moves by Black)**.** *In ordinary chess the piece inventory is finite, Black takes legal turns, and a checkmating end state responds to an opponent's play. In a generalized GBG or theorem universe, there may be* 10,000 *candidate theorems, or infinitely many; each gives a possible target for some task. These need not be interchangeable winning conditions for the* same *fixed task. A formal obstruction does not take a turn, and a newly proved lemma changes the prover's information rather than the truth or location of the fixed formula c. A fixed hidden password behaves similarly; only an expressly adaptive password protocol has a changing challenge.*

3 The first-person conjecture "White wins"

The phrase "White wins" has two distinct uses that must not be conflated: a chess *objective* (possibly a forced-win claim), and the proposed *self-settlement conjecture* of a specified machine.

Definition 3.1 (Bounded first-person settlement event)**.** *Freeze an identified machine instance M, theory T, target c, verifier V, admissible resources b, and any prescribed seeds or execution inputs ξ. Define*

$$\mathsf{Win}_b(M, T, c, V, \xi) = 1$$

iff the actual authenticated execution of M with ξ emits by budget b a complete π such that $V(\pi, T \vdash c) = 1$. The first-person sentence "I will prove the theorem" denotes this event for the agreed speaker M and protocol boundary.

This is neither (a) the imperative "prove c," (b) "I can prove c," nor (c) "there exists some legal successful continuation." Even a theorem establishing that a path *exists* is not the event that M *actually takes* it. The relevant subject must be identified: if an activated graduate federation proves c but the original M does not emit the required proof, the frozen first-person conjecture about M has not thereby succeeded.

Remark 3.2 (Time bounds and the meaning of failure)**.** *A trustworthy complete bounded execution that ends without an accepted proof refutes that* bounded future-event prediction*; it does not refute c. With no finite bound, nontermination or repeated failure is not, in general, a certificate that success will never occur. Nor does a plausible informal proof count as a machine-accepted certificate. This distinction is exemplified by the partial Metamath/*`aaliou`* continuation in [2].*

4 Player 0 and the recursive federation

The GBG+ reference [1] distinguishes the board-state data B from the board-player $P_0 = B^\star$. The minimal framework allows *only declared legal board actions*; memory, scheduling, certification, and federation are optional packages that must be explicitly adopted. This note proposes an *additional legal activation rule*, not a consequence automatically present in the minimal GBG+ axioms.

Definition 4.1 (Federation-activation transition)**.** *Let a complete enhanced state be*

$$S = (q, \mathcal{A}, \mathcal{F}, \mathsf{BANK}, \sigma, \Pi, \beta),$$

where q is the frozen object target, $\mathcal{A}$ lists active agents, $\mathcal{F}$ records frontiers and subfederations, BANK *is certified shared memory, σ the scheduler, Π provenance, and β charged remaining resources. A board-player activation is a partial rule*

$$\mathsf{Activate}_0(S, \mathit{target}, \mathit{budget}, \mathit{protocol}) \rightharpoonup S'$$

whose domain requires explicit authorization, adequate resources, a specified executable/federation identity, and preserved target/verifier invariants.

The parent can activate an AMLD+0 federation to explore a subgoal ℓ, or to investigate the first-person settlement conjecture "White wins." Graduate federations may be nested only where a recursion/activation rule expressly permits it. Activation, verification, routing, BANK access, and handoff are *charged*; unlogged solver substitutions and silent changes of the frozen target are prohibited.

Agent names. The historical AMLD+0 organization uses $0, P, R, D, I, C$ (Player 0, Proof, Refutation, Dependence, Independence, and an additional settlement/contradiction investigation role). The newer discussion also proposes K for a focused contradiction investigation and U for unresolved/unknown reporting. We *do not* silently replace C or redefine the original executable by introducing K, U; a seven-participant $0 + \{P, R, D, I, K, U\}$ configuration is a *proposed variant* until a specific protocol is approved.

5 Three distinct basin-escape methods

Definition 5.1 (A: self-settlement / "White wins" prompt)**.** *On detecting stagnation, Player 0 may legally assign an identified conjecture settler the first-person target*

$$\mathsf{SelfWin}(M, q, b): \quad \textit{"I will prove the theorem"}$$

(or "I will reach the target" in GBG terms). A successful operational test requires the designated M to reach its declared target and emit the accepted certificate, not merely to predict that it will do so.

Definition 5.2 (B: Eureka-style lemma fishing)**.** *Given hypotheses H and target c, a Eureka prompt seeks a nontrivial family $L = \{\ell_1, \ldots, \ell_k\}$ with accepted certificates for*

$$H \vdash \ell_i \quad (1 \le i \le k), \qquad H \cup L \vdash c.$$

The intended discovery must not merely rename the target or smuggle in the completed proof; it should yield reusable intermediate certificates that bridge a search obstacle.

Although $H \vdash \ell_i$ means $H \cup L$ has no new logical consequences beyond H, using the *right* proven intermediary can dramatically affect search order and proof discovery cost. Its usefulness is computational, not the addition of a new axiom.

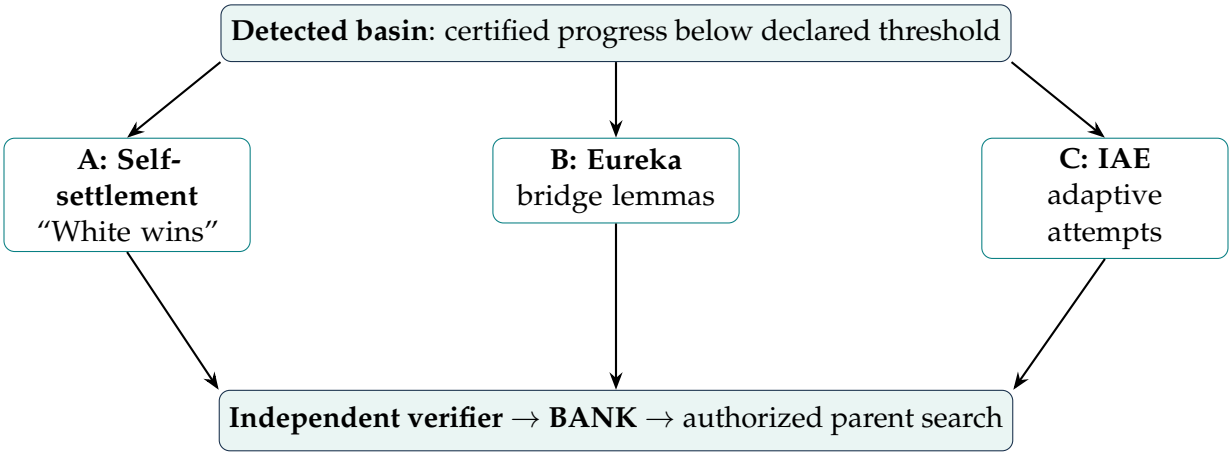
The Eureka search space may draw from an infinite candidate family $\mathcal{T}$, while any particular finite certificate or finite BANK handoff uses only finitely many derivation steps and appropriately typed

hypotheses. This allows Player 0 to redirect search toward a useful intermediate target t_i *without replacing* the frozen conjecture c. The number of possible intermediate targets, and the number of possible obstructions, may both be infinite; no completeness, finite-time discovery, or guaranteed escape follows from that abundance.

Definition 5.3 (C: Iterative Adaptive Eureka (IAE))**.** *IAE is a budgeted sequence of Eureka-oriented attempts with legal adaptive updates:*

$$(h_0, \mathsf{BANK}_0) \to (h_1, \mathsf{BANK}_1) \to \cdots \to (h_m, \mathsf{BANK}_m),$$

where h_t records a prompt/search policy, an update depends on allowed observed feedback, and only independently checked results enter BANK_{t+1}. Stagnation rules, iteration caps, verification charges, and stopping conditions are explicit protocol parameters, not assumed guarantees.



The three methods are different *interventions*, not disjoint logical truth values: A changes the explicit *meta-target*; B searches for an *intermediate theorem*; C changes the *search policy over successive attempts*. IAE can itself invoke Eureka repeatedly, and a self-settlement federation may use either method. None guarantees escape.

6 BANK: reusable knowledge without illicit hypotheses

Definition 6.1 (Trusted BANK entry)**.** *A trusted BANK entry is a tuple $e = (\ell, \pi, \Gamma, \mathit{origin}, \mathit{cost})$ where the verifier accepts that the derivation π establishes $\Gamma \vdash \ell$, where Γ records all undischarged assumptions. Unverified suggestions belong in a separate proposal store (such as a Futurebank) and may not be used as certified unconditional facts.*

Lemma 6.2 (No hypothesis laundering)**.** *If the verifier is sound and BANK faithfully retains each certificate's undischarged hypothesis set, sharing a BANK entry cannot turn a conditional theorem $T \cup \Delta \vdash \ell$ into an unconditional theorem $T \vdash \ell$ without a further valid discharge of Δ.*

Proof. Certificate acceptance authorizes precisely the stated sequent. By assumption, any later use must preserve its dependencies or provide a verifier-accepted derivation discharging them. A use that omits Δ without discharge is not a legal BANK inference. □

A shared BANK may contain proof certificates, countermodels or model evidence with their exact logical status, reachability certificates, and machine-checkable dependency records. "All lemmas" means all *verified and legally eligible* lemmas, not all generated claims. Priorities and indices may guide retrieval while retaining the full archive.

7 Six investigation reports are not six mutually exclusive truth values

Proposed investigations include: P (prove c), R (prove $\neg c$), D (investigate dependence on recorded assumptions), I (certify independence relative to a specified theory), K (derive $c \to \bot$), and U (report no determination within a stated window). These are useful *work streams*, but not automatically a disjoint six-valued logic.

Proposition 7.1 (Overlap under classical logic)**.** *For a consistent classical base theory with ordinary negation,*

$$T \vdash (c \to \bot) \quad \Longleftrightarrow \quad T \vdash \neg c.$$

Thus positive reports from K and R can certify the same logical outcome by different search routes.

Proof. In classical formalizations $\neg c$ may be defined as $c \to \bot$, or is interderivable with it. □

A dependence report requires its own exact property (e.g. $T \cup \Delta \vdash c$, a necessary hypothesis, or a relative consistency claim). Independence of c from T means $T \nvdash c$ and $T \nvdash \neg c$ and generally needs a valid metamathematical certificate, not just a long unsuccessful search. U is a bounded *unknown report*, not a certificate of independence. In an inconsistent theory both directions may be derivable, which also rules out naive mutual exclusivity.

8 The SIC preorder and settlement regions

Let $\mathscr{S}$ be a class of admissible verifier-gated simulation systems. Write $X \preceq Y$ if Y can reproduce the relevant certified behavior of X through a specified legal simulation, including faithful certificate checking and (if a bound is imposed) a declared resource-accounting transformation. Assume $\preceq$ is reflexive and transitive. Quotient by mutual simulation $X \sim Y \iff X \preceq Y \wedge Y \preceq X$ to obtain the poset $\mathcal{P} = \mathscr{S}/\!\sim$.

Definition 8.1 (Certificate settlement region)**.** *For a fixed objective (T, c, V) and a specified class of admissible protocols, set*

$$U_c = \{[X] \in \mathcal{P} : X \textit{ emits an accepted certificate of } T \vdash c \textit{ or } T \vdash \neg c\}.$$

This is object-level *settlement. For the first-person prediction about M, use a separate region $U_{\mathsf{Win}_b(M,T,c,V,\xi)}$ with an explicitly fixed subject and event-verification semantics; the two regions are not interchangeable.*

Theorem 8.2 (Upward closure, conditional on certificate preservation)**.** *If $X \preceq Y$ guarantees that every accepted settlement certificate produced by X can be legally reproduced or transferred to Y under the applicable cost regime, then U_c is an upper set:*

$$[X] \in U_c, \quad [X] \preceq [Y] \quad \Longrightarrow \quad [Y] \in U_c.$$

Proof. Choose an accepted certificate produced by X. The stated simulation property makes it reproducible or transferable to Y without invalidating its certificate or the resource rules. Therefore Y settles the same fixed objective. □

Definition 8.3 (Settlement spectrum)**.** *When minimal elements exist, define* $\mathcal{M}_c = \mathrm{Min}(U_c)$. *This records minimal* simulation types*, not just a scalar count of proof steps.*

Theorem 8.4 (Minimal-antichain generation)**.** *If* $\mathcal{P}$ *is well-founded, every upper set* $U \subseteq \mathcal{P}$ *satisfies*

$$U = {\uparrow}\,\mathrm{Min}(U),$$

and $\mathrm{Min}(U)$ *is an antichain. In particular this applies to* U_c.

Proof. For $x \in U$, the nonempty subset $U \cap {\downarrow}\, x$ has a minimal element m by well-foundedness. This m is minimal in U, and $m \preceq x$. Therefore $x \in {\uparrow}\,\mathrm{Min}(U)$. The opposite inclusion follows from upward closure. Distinct minimal elements are incomparable. □

Remark 8.5 (Essential repairs to overstrong claims)**.** *Without well-foundedness an upper set may have* no *minimal elements:* $\{q \in \mathbb{Q} : q > 0\}$ *is an upper set of the usual rational chain with none. It also shows a totally ordered settlement region need not be of the closed form* $[d, \infty)$*; it may be open at its lower cut. A threshold theorem of that form is justified for a well-ordered discrete depth scale, not for arbitrary chains. Empty* $\mathcal{M}_c$ *does not by itself prove logical independence unless minimal elements are known to exist whenever the region is nonempty.*

Definition 8.6 (Derived conjecture preorder)**.** *For object-level tasks with a common settledness semantics, define*

$$c \leq_{\mathrm{SIC}} d \quad \Longleftrightarrow \quad U_d \subseteq U_c.$$

This is a preorder (reflexive, transitive). It orders conjectures by the inclusion of their settlement-capability regions, not necessarily by elapsed time or proof length.

Proposition 8.7 (Negation symmetry of object-level settlement)**.** *In classical logic, suppose that accepted proof certificates may be converted between* $T \vdash \neg\neg c$ *and* $T \vdash c$ *within the applicable resource regime, and that both settlement directions have the same admissible simulation class. Then* $U_c = U_{\neg c}$. *Consequently, where defined,* $\mathcal{M}_c = \mathcal{M}_{\neg c}$.

Proof. Settling c requires a proof of c or $\neg c$; settling $\neg c$ requires a proof of $\neg c$ or $\neg\neg c$. Classical double-negation conversion supplies the corresponding certified derivation, by the assumed resource-admissible conversion. □

Important qualification. The symmetry does *not* say that the personal prediction "I will prove c" is interchangeable with "I will prove $\neg c$." Those are different events; only the symmetric *object-level* task "settle one of the two directions" has this property.

9 Federation and cost: reachability need not mean speed

Let A_0 be a legal action system without graduate activation and A_1 an extension with Player-0 activation, BANK sharing, Eureka, and IAE. Assume every old legal behavior remains available unchanged and is still verified by the same V.

Proposition 9.1 (Weak monotonicity of feasible paths)**.** *If* $\mathsf{A}_0 \subseteq \mathsf{A}_1$ *in the above operational sense, every settlement path feasible in* A_0 *is still feasible in* A_1 when no additional mandatory overhead is imposed. *Thus optional new actions cannot remove previously reachable certificates.*

Proof. A valid old action path remains legal and uses the old transitions unchanged. □

This is *not* a theorem that the new configuration is faster. A forced activation tax, expensive certificate transport, BANK-retrieval costs, contention, repeated discoveries, or poor scheduling can make a federation slower. For example, a one-step baseline proof with an obligatory ten-step activation is slower in the new configuration. A meaningful speedup compares fully charged expected settlement costs and, separately, wall-clock times.

Research conjecture 9.2 (Cooperative basin escape)**.** *There exist nontrivial families of frozen proof tasks and specified search budgets for which a legally activated heterogeneous federation with verifier-gated BANK sharing, and a Player-0 policy choosing among self-settlement, Eureka, and IAE, has lower expected* fully charged *settlement cost than a matched solo baseline and a matched nonsharing multi-agent baseline.*

This is a proposed research conjecture, not a proven universal theorem. In particular, producing an elegant intermediate lemma need not help a given solver discover a complete formal certificate.

10 Protocol sketch for an eventual controlled test

The following is a design specification, *not* an initiated experiment.

1. **Freeze the task.** Record T, c, V, executable hashes, seeds, legal rules, deadline/budget, source restrictions, and the exact referent of "I." In chess, fix admissible starting states and whether the target means actual win or forced strategy. In password trials, state whether the secret can change.
2. **Detect quicksand without inventing refutation.** Use a specified window and measurable indicators: new accepted BANK entries, discharged obligations, verified frontier contraction, accepted witnesses per charged computation, or other frozen metrics.
3. **Prespecify the alternatives.** Compare (i) one prover, (ii) matched independent workers with no shared BANK, (iii) shared BANK without adaptive activation, and (iv) Player-0-activated federation with A/B/C escape policies. Allocate equal total charged resource budgets and also record wall-clock outcomes.
4. **Log legal activation and provenance.** Every promotion into BANK records source, support obligations, proof certificate, verifier result, and charge. No unauthorized engine substitution, target change, or unlogged hidden oracle is permitted.
5. **Keep findings distinct.** Report object-level `PROVED`/`REFUTED`/suitably certified `INDEPENDENT`, conditional or dependence findings, and bounded `UNKNOWN`. Record the separate truth of each first-person future-event claim.
6. **Audit before publication.** Re-run accepted certificates through the frozen independent checker and publish the ledger and comparator definitions. A plausible derivation is not a completed machine proof.

11 Summary and precise research question

The analogy "I am unsure how to move in chess" captures *search uncertainty*. It does not license a direct transfer of chess's adversarial quantifiers to static theorem proving. Black moves in chess; the formal target stays fixed in ordinary ATP even when the target family and obstruction family are infinite; a password can be fixed or can evolve if the protocol says so. A GBG+ board may be *equipped* with legal Player-0 activation, allowing graduate AMLD+0 federations to investigate either the first-person settlement event or intermediate proof obligations. Eureka, IAE, and "White wins" provide three distinct but composable escape interventions. BANK unites their certified contributions. The SIC preorder then asks which simulation capabilities, alone or in a federation, suffice for settlement and how minimal capability types compare.

Primary open question. Can a Player-0-governed federation transform a stalled *search policy* into a verifier-accepted settlement by discovering and sharing a certificate-bearing bridge, at demonstrably lower fully charged cost than matched alternatives—without changing the frozen conjecture or confusing a first-person prediction with a command?

Source materials and provenance

[1] Brian Tenneson, *Generalized Board Games and GBG+: A Self-Contained Formal Foundation with the Board as a Player*, version 0.04, 26 September 2026, esp. GBG rule-complete states; Player-0 authorization; optional certification, memory, federation and scheduler packages; canonical ATP specialization. `https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/gbg_and_gbg_plus_formal_foundations_004/`.

[2] *DATA MIND's Frontiers of Discovery, Volume I: The Quicksand Problem*, working illustrated manuscript, 16 October 2026, especially the Grass-Growth Principle and the distinction between an informal mathematical bridge and a formally certified operational result. Supplied in this conversation as `DATA_MIND_Volume_1_Quicksand(2).pdf`.

[3] *DATA'S MIND Through Chess: Location Reports, Strategy Trees, and Verifier-Gated Theorem Settling*, supplied in this conversation as `DATAs_Mind_Chess_Introduction(10).pdf`, esp. the difference between a favorable path and a forced-winning strategy tree.

[4] *Conjecture Settling Part I*, 11-page working note supplied in this conversation as `conjecture_settling(1).pdf`; source of the SIC settlement-region/preorder discussion. The present note states corrected conditional versions where needed.

Provenance / status. This document formalizes proposals discussed with the researcher. Theorems above are elementary conditional mathematical consequences of the definitions and stated assumptions; there has been no automated AMLD+0 run, chess engine substitution, proof-verifier experiment, or external independent peer audit in preparing this note. Nothing here changes a frozen experimental protocol or authorizes its execution.